I joined John Wheeler's research group as a graduate student in 1962, shortly after Khalatnikov and Lifshitz had published their calculation, and shortly after Lifshitz together with Landau had enshrined the calculation and its "no singularity" conclusion in a famous textbook, *The Classical Theory of Fields*. I recall vividly Wheeler encouraging his research group to study the calculation. If it is right, its consequences are profound, he told us. Unfortunately, the calculation was extremely long and complicated, and the published details were too sketchy to permit us to check them-and Khalatnikov and Lifshitz were confined within the Soviet Union's iron curtain, so we could not sit down with them and discuss the details.

Nevertheless, we began to contemplate the possibility that the imploding Universe, upon reaching some very small size, *might* "bounce" and reexplode in a new "big bang," and similarly that an imploding star, after sinking inside its horizon, *might* bounce and reexplode.

But where could the star go if it reexplodes? It surely could not explode back out through the hole's horizon. Einstein's laws of gravity forbid anything (except virtual particles) to fly out of the horizon. There was another possibility, however: *The star might manage to explode into some other region of our Universe, or even into another universe.*

Figure 13.4 depicts such an implosion and reexplosion using a sequence of embedding diagrams. (Embedding diagrams, which are quite different from spacetime diagrams, were introduced in Figures 3.2 and 3.3.)

Each diagram in Figure 13.4 depicts our Universe's curved space, and the curved space of another universe, as two-dimensional surfaces embedded in a higher-dimensional *hyperspace*. [Recall that hyperspace is a figment of the physicists' imagination: We, as humans, are confined always to live in the space of our own Universe (or, if we can get there, the space of the other universe); we can never get out of those spaces into the surrounding higher-dimensional hyperspace, nor can we ever receive signals or information from hyperspace. The hyper-space serves only as an aid in visualizing the curvature of space around the imploding star and its black hole, and in visualizing the manner in which the star can implode in our Universe and then reexplode into another universe.]

In Figure 13.4, the two universes are like separate islands in an ocean and the hyperspace is like the ocean's water. Just as there is no land